variety. But just as the surface of the tightrope has an additional, small, curled-up, circular dimension, maybe the fabric of space also has a small, curled-up, circular dimension, one so small that no one has powerful enough magnifying equipment to reveal its existence. Because of its tiny size, Klein argued, the dimension would be hidden.

How small is small? Well, by incorporating certain features of quantum mechanics into Kaluza's original proposal, Klein's mathematical analysis revealed that the radius of an extra circular spatial dimension would likely be roughly the Planck length,[16] certainly way too small for experimental accessibility (current state-of-the-art equipment cannot resolve anything smaller than about a thousandth the size of an atomic nucleus, falling short of the Planck length by more than a factor of a million billion). Yet, to an imaginary, Planck-sized worm, this tiny, curled-up circular dimension would provide a new direction in which it could roam just as freely as an ordinary worm negotiates the circular dimension of the tightrope in Figure 12.5. Of course, just as an ordinary worm finds that there isn't much room to explore in the clockwise direction before it finds itself back at its starting point, a Planck-sized worm slithering along a curled-up dimension of space would also repeatedly circle back to its starting point. But aside from the length of the travel it permitted, a curled-up dimension would provide a direction in which the tiny worm could move just as easily as it does in the three familiar unfurled dimensions.

To get an intuitive sense of what this looks like, notice that what we've been referring to as the tightrope's curled-up dimension—the clockwise/ counterclockwise direction—*exists at each point along its extended dimension.*The earthworm can slither around the circular girth of the tightrope at any point along its outstretched length, and so the tightrope's surface can be described as having one long dimension, with a tiny, circular direction tacked on at each point, as in Figure 12.6. This is a useful image to have in mind because it also applies to Klein's proposal for hiding Kaluza's extra dimension of space.

To see this, let's again examine the fabric of space by sequentially showing its structure on ever smaller distance scales, as in Figure 12.7. At the first few levels of magnification, nothing new is revealed: the fabric of space still appears three-dimensional (which, as usual, we